

Community+ Economic Control Mechanism (ECM)

Formal Whitepaper

Abstract

This paper presents the Community+ Economic Control Mechanism (ECM), a sustainability-first token-economic framework designed to automatically regulate token supply based on real-world platform economics. The ECM links token emissions and withdrawals directly to the relationship between operating costs and earned revenue, ensuring long-term platform viability, preventing speculative excess, and aligning contributor incentives with financial reality.

Design Principles

The ECM is founded on revenue-backed emissions, automatic austerity, predictability over volatility, utility-first design, and regulatory prudence.

Economic Variables

C_t represents total operating cost per period.

R_t represents total platform revenue per period.

Economic Coverage Ratio

$$ECR_t = R_t / C_t$$

ECR below 1.0 indicates deficit, equal to 1.0 indicates equilibrium, and above 1.0 indicates surplus.

Equilibrium Band

A control band of 0.95 to 1.10 is defined to prevent oscillation.

Token Emission Function

$$\text{Emission}_t = \min(T_{\text{max}}, k_e \times (ECR_t - 1))$$

Token Sink Function

$$\text{Sink}_t = \min(T_{\text{smax}}, k_s \times (1 - ECR_t))$$

Treasury Dynamics

The treasury acts as the buffer and counterparty to all token flows.

Conclusion

Community+ emits tokens only when real revenue exceeds real cost and absorbs tokens when it does not.